Practice Problems

Instructions: Take input from the user. Use the loop or slicing technique specified in each question. Match the sample output format. Solutions are not included.

Problem 1

Take an integer. Print whether it is positive, negative, or zero; whether it is even or odd (if nonzero); and its square.

Sample Input: -6

> Sample Output:
> Negative Number
> Even Number
> Square: 36

Problem 2

Take a positive integer n. Using a for loop, find the sum and count of even numbers and odd numbers from 1 through n.

Sample Input: 10

> Sample Output:
> Even Sum: 30
> Odd Sum: 25
> Even Count: 5
> Odd Count: 5

Problem 3

Take an integer n and print its multiplication table from 1 through 10 using a for loop.

Sample Input: 5

> Sample Output:
> 5 x 1 = 5
> 5 x 2 = 10
> ...
> 5 x 10 = 50

Problem 4

Take a nonnegative integer n. Calculate n! using a while loop. Assume 0! = 1.

Sample Input: 5

> Sample Output:
> Factorial: 120

Problem 5

Take a positive integer. Using a while loop, count its digits, calculate the sum of its digits, and print its last digit.

Sample Input: 4532

Sample Output:
Total Digits: 4
Sum of Digits: 14
Last Digit: 2

Problem 6

Take marks of 6 subjects and store them in a list. Print the complete list, first 3 and last 3 marks using slicing, total, average (2 decimal places), and count of subjects with marks ≥ 50.

Sample Input: 75, 40, 85, 60, 45, 90

Sample Output:
First 3: [75, 40, 85]
Last 3: [60, 45, 90]
Total: 395
Average: 65.83
Passed Subjects: 4

Problem 7

Take 8 integers into a list. Using a loop and if-else statements, count positive numbers, negative numbers, and zeros.

Sample Input: 4, -2, 0, 7, -5, 0, 9, -1

Sample Output:
Positive: 3
Negative: 3
Zero: 2

Problem 8

Take a list of 8 integers. Using slicing, print the first 4 elements, last 4 elements, every second element starting at index 0, and the reversed list.

Sample Input: 10, 20, 30, 40, 50, 60, 70, 80

Sample Output:
First 4: [10, 20, 30, 40]
Last 4: [50, 60, 70, 80]
Alternate: [10, 30, 50, 70]
Reversed: [80, 70, 60, 50, 40, 30, 20, 10]

Problem 9

Take 5 integers into a list. Using a for loop (without max() or min()), find the largest, smallest, and their difference.

Sample Input: 12, 45, 7, 30, 18

Sample Output:
Largest: 45
Smallest: 7
Difference: 38

Problem 10

Take a string containing only English letters. Using a loop, count vowels (a, e, i, o, u) and consonants; print the reversed string using slicing.

Sample Input: Programming

Sample Output:
Vowels: 3
Consonants: 8
Reversed: gnimmargorP

Problem 11

Take a positive integer n. Use nested for loops to print a right-angled star triangle with n rows.

Sample Input: 5

Sample Output:
```
*
**
***
****
*****
```

Problem 12

Take a positive integer n. Use nested loops to print an inverted star triangle with n rows.

Sample Input: 5

Sample Output:
```
*****
****
***
**
*
```

Problem 13

Take a positive integer n. Use nested loops to print increasing sequences from 1 to the row number.

Sample Input: 5

Sample Output:
```
1
12
123
1234
12345
```

Problem 14

Take a positive integer n. Use nested loops to print each row number repeated that many times.

Sample Input: 5

Sample Output:
1
22
333
4444
55555

Problem 15

Take the number of products n, then input each product price using a loop and store prices in a list. Calculate the total using a loop and count prices greater than 500 taka. Apply discounts: total below 1000 → 0%; 1000 to below 3000 → 5%; 3000 or above → 10%. Print total, count, discount amount, and final bill.

Sample Input: Number of Products: 4
Prices: 500, 800, 1200, 600

Sample Output:
Total Bill: 3100
Products Above 500: 3
Discount: 310
Final Bill: 2790